GRAPHS WITH $\lfloor n^2/4 \rfloor + 1$ EDGES CONTAIN AT LEAST $\lfloor n/2 \rfloor$ TRIANGLES: A LEAN 4 PROOF OF RADEMACHER'S THEOREM

MATH0

Abstract. We give a complete, self-contained proof of Rademacher's theorem: every simple graph on n vertices with exactly $\lfloor n^2/4 \rfloor + 1$ edges contains at least $\lfloor n/2 \rfloor$ triangles. The proof is a strong induction over vertex subsets. It derives Mantel's theorem along the way and splits on the degree sums along the edges of a triangle. The result is classical. It is the case $t = 1$ of Erdős Problem #1010, whose general form was proved by Lovász–Simonovits and by Nikiforov–Khadzhiivanov, and a Lean proof of that general form already exists. Our contribution is therefore an independent formal proof of a known result. It is verified in Lean 4 with Mathlib against the statement `Erdos1010.erdos_1010.variants.rademacher` of the formal-conjectures repository and uses only the standard axioms.

1. Introduction

Mantel's theorem [8] says that a graph on n vertices with more than $\lfloor n^2/4 \rfloor$ edges contains a triangle. The bound is attained by the balanced complete bipartite graph $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$. In 1941 Rademacher proved, without publishing it, that one edge beyond this threshold already forces many triangles: every graph on n vertices with $\lfloor n^2/4 \rfloor + 1$ edges contains at least $\lfloor n/2 \rfloor$ triangles. The result first appeared in print, with a short proof, in a paper of Erdős [2]; see also [3] and the account in [5]. The bound is sharp. Add one edge inside the larger part of $K_{\lceil n/2 \rceil, \lfloor n/2 \rfloor}$ (with $n \geq 3$). Every triangle of the resulting graph contains the added edge, and its third vertex lies in the other part, so there are exactly $\lfloor n/2 \rfloor$ triangles.

Erdős [3] extended Rademacher's theorem: a graph on n vertices with $\lfloor n^2/4 \rfloor + t$ edges contains at least $t\lfloor n/2 \rfloor$ triangles whenever $t < cn$, for some absolute constant $c > 0$. He asked whether the same holds for all $t < \lfloor n/2 \rfloor$. This is Erdős Problem #1010 on the website maintained by Bloom [1]. The question was answered affirmatively, independently by Lovász and Simonovits [7] and by Nikiforov and Khadzhiivanov [11]. The more general Erdős–Rademacher problem asks for the minimum number of triangles in a graph of given order and size. It has a large literature. For large n and edge density bounded away from 1, it was solved exactly by Liu, Pikhurko and Staden [6], whose introduction surveys the history.

Formal statements and prior formalization. The formal-conjectures repository [4] formalizes Problem #1010 in Lean 4 as `Erdos1010.erdos_1010`. It also records Rademacher's case $t = 1$ as a separate statement, `Erdos1010.erdos_1010.variants.rademacher`, which is the statement proved here. Both are tagged `research solved`. The upstream file attaches a `formal_proof` link to the general statement `erdos_1010`. The link points to a Lean 4 proof in the repository `plby/lean-proofs` [12] (file `src/latest/ErdosProblems/Erdos1010.lean`, commit `8822f7d`). That proof covers every finite vertex type and every $t < \lfloor n/2 \rfloor$. Taking $t = 1$ gives the present statement for $n \geq 4$. The remaining cases are trivial (Remark 2). The variant itself carries no `formal_proof` attribute upstream, but a formal proof of a statement that implies it was already available. We therefore do not claim a first formalization.

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What this note adds is an independent machine-checked proof of the variant, stated exactly as upstream. The proof is a short, self-contained double induction and needs about 330 lines of Lean. According to its header, the existing development uses maximum-cut charge bounds and a reduction for odd orders. The argument formalized here is elementary, and we make no claim that it is new as a piece of mathematics. Its first half is the standard inductive proof of Mantel's theorem.

2. Statement

All graphs are finite and simple. For a graph G we write $u \sim v$ when u and v are adjacent, $e(G)$ for the number of edges, and $t(G)$ for the number of triangles, i.e. of 3-element vertex sets that span a complete graph.

Theorem 1 (Rademacher)**.** *Let $n \geq 0$ and let G be a simple graph on the vertex set $\{0, \dots, n-1\}$ with $e(G) = \lfloor n^2/4 \rfloor + 1$. Then $t(G) \geq \lfloor n/2 \rfloor$.*

The upstream statement is copied below verbatim from the file `FormalConjectures/ErdosProblems/1010.lean` of formal-conjectures at commit `c77fdd4`. It sits inside `namespace Erdos1010`, with `open SimpleGraph` in effect:

```
open scoped Classical in
/--
Rademacher proved that every graph on $n$ vertices with $\lfloor n^2/4\rfloor+1$ edges contains at
least $\lfloor n/2\rfloor$ triangles.
-/
@[category research solved, AMS 5]
theorem erdos_1010.variants.rademacher : ∀ n : ℕ, ∀ G : SimpleGraph (Fin n),
    G.edgeFinset.card = n ^ 2 / 4 + 1 → n / 2 ≤ (G.cliqueFinset 3).card := by
  sorry
```

Here `n ^ 2 / 4` and `n / 2` use natural-number division, so they equal $\lfloor n^2/4 \rfloor$ and $\lfloor n/2 \rfloor$. The finset `G.cliqueFinset 3` consists of the 3-element vertex sets that are cliques, i.e. the unordered triangles. The decidability instances that `edgeFinset` and `cliqueFinset` need come from `open scoped Classical`. So the formal statement is exactly Theorem 1.

Remark 2. For $n \leq 2$ the hypothesis is impossible, because $e(G) \leq \binom{n}{2} < \lfloor n^2/4 \rfloor + 1$. For $n = 3$ it forces $G = K_3$, and then $t(G) = 1 = \lfloor 3/2 \rfloor$. The proof below does not treat these cases separately. The theorem also holds with "at least $\lfloor n^2/4 \rfloor + 1$ edges", since deleting edges cannot create triangles. In fact the Lean proof establishes the "at least" form for every vertex subset (Theorem 4).

3. Proof

Fix a finite vertex set V and a simple graph G on V. All sets below are subsets of V.

Definition 3. For $S \subseteq V$ and $v \in V$ (not necessarily in S) let $[v \sim w] \in \{0, 1\}$ be the adjacency indicator, and define

$$d_S(v) = \sum_{w \in S} [v \sim w], \qquad D(S) = \sum_{v \in S} d_S(v), \qquad \mathcal{T}(S) = \{U \subseteq S : \ U \text{ is a triangle of } G\}.$$

So $d_S(v) = |N(v) \cap S|$ and $D(S) = 2e(G[S])$. In Lean these are `dg`, `D` and `tri`; `tri G S` is `(G.cliqueFinset 3).filter (· ⊆ S)`.

We prove the following stronger statement. It is the Lean lemma `main`.

Theorem 4. *For every $S \subseteq V$, if $D(S) \geq 2\big(\lfloor |S|^2/4 \rfloor + 1\big)$, then $|\mathcal{T}(S)| \geq \lfloor |S|/2 \rfloor$.*

3.1. **Bookkeeping lemmas.**

Lemma 5 (`dg_erase`, `D_erase`)**.** *Let $x \in S$. For every $v \in V$, $d_S(v) = [v \sim x] + d_{S\setminus x}(v)$. Moreover $D(S) = D(S \setminus x) + 2d_S(x)$.*

Proof. The first identity splits off the term $w = x$. For the second, split off $v = x$ in $D(S)$ and apply the first identity to the remaining terms:

$$D(S) = d_S(x) + \sum_{v \in S\setminus x} \big([x \sim v] + d_{S\setminus x}(v)\big) = d_S(x) + d_{S\setminus x}(x) + D(S \setminus x).$$

Here we used the symmetry $[v \sim x] = [x \sim v]$. Finally $d_{S\setminus x}(x) = d_S(x)$, because $[x \sim x] = 0$. □

Lemma 6 (`D_erase_two`)**.** *If $x, y \in S$ and $x \sim y$, then $D(S)+2 = D(S\setminus\{x,y\})+2d_S(x)+2d_S(y)$.*

Proof. We have $y \in S \setminus x$ since $x \neq y$. Apply Lemma 5 twice: $D(S) = D(S \setminus \{x,y\}) + 2d_{S\setminus x}(y) + 2d_S(x)$. Also $d_S(y) = [y \sim x] + d_{S\setminus x}(y) = 1 + d_{S\setminus x}(y)$. □

Lemma 7 (`sq_div_four`)**.** *For every $M \in \mathbb{N}$, $\lfloor (M+2)^2/4 \rfloor = \lfloor M^2/4 \rfloor + M + 1$.*

Proof. $(M+2)^2 = M^2 + 4(M+1)$. □

Lemma 8 (`step_bound`)**.** *Let $x, y \in S$ with $x \sim y$ and $d_S(x) + d_S(y) \leq |S|$, and suppose $D(S) \geq 2(\lfloor |S|^2/4 \rfloor + 1)$. Put $S' = S \setminus \{x,y\}$. Then $|S'| = |S| - 2$ and $D(S') \geq 2(\lfloor |S'|^2/4 \rfloor + 1)$.*

Proof. Let $M = |S'|$, so $|S| = M + 2$. By Lemmas 6 and 7,

$$D(S') = D(S)+2-2\big(d_S(x)+d_S(y)\big) \geq 2\big(\lfloor M^2/4 \rfloor + M + 2\big) + 2 - 2(M+2) = 2\big(\lfloor M^2/4 \rfloor + 1\big).$$ □

Lemma 9 (`exists_common`)**.** *If $x, y \in V$ and $d_S(x) + d_S(y) > |S|$, then some $z \in S$ satisfies $x \sim z$ and $y \sim z$.*

Proof. The sets $N(x) \cap S$ and $N(y) \cap S$ are subsets of S whose sizes add up to more than $|S|$. By inclusion–exclusion their intersection is nonempty. □

3.2. **Mantel's theorem.**

Lemma 10 (Mantel; `mantel`)**.** *If $D(S) \geq 2(\lfloor |S|^2/4 \rfloor + 1)$, then there are $x, y, z \in S$ with $x \sim y$, $x \sim z$ and $y \sim z$.*

Proof. We use strong induction on $|S|$. Since $D(S) > 0$, S contains an edge xy (`exists_adj_of_D_pos`). If $d_S(x) + d_S(y) > |S|$, Lemma 9 gives a common neighbour $z \in S$. Otherwise Lemma 8 shows that $S \setminus \{x,y\}$ satisfies the hypothesis. Since it is smaller than S, the induction hypothesis gives a triangle inside it, and hence inside S. □

The three vertices are automatically distinct, because G has no loops. So $\{x,y,z\} \in \mathcal{T}(S)$ (`mem_tri`).

3.3. **The two cases.** Let $N = |S|$ and let xyz be a triangle in S.

Lemma 11 (Case 1; `case_one`)**.** *Suppose $D(S) \geq 2(\lfloor N^2/4 \rfloor + 1)$ and $d_S(x) + d_S(y) \leq N$. Put $S' = S \setminus \{x,y\}$ and suppose Theorem 4 holds for S'. Then $|\mathcal{T}(S)| \geq \lfloor N/2 \rfloor$.*

Proof. By Lemma 8, S' satisfies the hypothesis of Theorem 4 and $|S'| = N - 2$, so $|\mathcal{T}(S')| \geq \lfloor (N-2)/2 \rfloor$. Every triangle in S' is a triangle in S (`tri_mono`). The triangle $\{x,y,z\}$ lies in $\mathcal{T}(S)$ but not in $\mathcal{T}(S')$, because $x \notin S'$. Hence $|\mathcal{T}(S)| \geq \lfloor (N-2)/2 \rfloor + 1 = \lfloor N/2 \rfloor$. The last equality uses $N \geq 2$, which holds because S contains the triangle $\{x,y,z\}$. □

Lemma 12 (Case 2; `case_two`)**.** *Suppose $d_S(x) + d_S(y) > N$, $d_S(y) + d_S(z) > N$ and $d_S(x) + d_S(z) > N$. Then $|\mathcal{T}(S)| \geq \lfloor N/2 \rfloor$.*

No assumption on the number of edges is needed here.

Proof. Let $R = S \setminus \{x, y, z\}$, so $|R| = N - 3$. Each of x, y, z is adjacent to the other two and not to itself. So three applications of Lemma 5 give

$$d_S(x) = 2 + d_R(x), \qquad d_S(y) = 2 + d_R(y), \qquad d_S(z) = 2 + d_R(z). \tag{1}$$

Index the three edges of the triangle by $i \in \{0, 1, 2\}$ as $e_0 = \{x, y\}$, $e_1 = \{y, z\}$, $e_2 = \{x, z\}$. In Lean, $e_i = \{p_i, r_i\}$ with $p = (x, y, x)$ and $r = (y, z, z)$. Let

$$P = \{(w, i) \in R \times \{0, 1, 2\} : \ w \text{ is adjacent to both endpoints of } e_i\}, \qquad f(w, i) = e_i \cup \{w\}.$$

Counting P. Fix $w \in R$ and let $k_w = [x \sim w] + [y \sim w] + [z \sim w] \in \{0, 1, 2, 3\}$. The three edges e_i are the three pairs of vertices from $\{x, y, z\}$. So the number of i with $(w, i) \in P$ equals $\binom{k_w}{2}$, and $k_w \le \binom{k_w}{2} + 1$ for $k_w \le 3$. Summing over $w \in R$ gives

$$d_R(x) + d_R(y) + d_R(z) \le |P| + |R|. \tag{2}$$

In Lean this is `sum_le_sum` followed by a case split over the eight adjacency patterns of w with x, y, z.

Injectivity. Each $f(w, i)$ is a triangle in S. Its vertices lie in S, e_i is an edge, and w is adjacent to both ends of e_i. The map f is injective on P. Indeed, suppose $f(w, i) = f(w', i')$. Then w is the only element of $f(w, i)$ outside $\{x, y, z\}$, so $w = w'$, and then $e_i = f(w, i) \cap \{x, y, z\}$ determines i. In Lean, $w = w'$ is shown first, and then all nine pairs (i, i') are checked by comparing which of x, y, z belong to the two sets. Finally $\{x, y, z\} \notin f(P)$, because every element of $f(P)$ contains a vertex of R. Therefore $\{\{x, y, z\}\} \cup f(P) \subseteq \mathcal{T}(S)$ is a disjoint union, and

$$|\mathcal{T}(S)| \ge |P| + 1.$$

Conclusion. By (2) and (1),

$$|\mathcal{T}(S)| \ge d_R(x) + d_R(y) + d_R(z) - |R| + 1 = d_S(x) + d_S(y) + d_S(z) - 6 - (N - 3) + 1.$$

Adding the three hypotheses gives $2(d_S(x) + d_S(y) + d_S(z)) \ge 3N + 3$. So $2|\mathcal{T}(S)| \ge 3N + 3 - 2N - 4 = N - 1$, and since $|\mathcal{T}(S)|$ is an integer, $|\mathcal{T}(S)| \ge \lceil (N - 1)/2 \rceil = \lfloor N/2 \rfloor$. In Lean this last step is a single `omega` call over natural numbers, using the facts above. □

3.4. **Proof of Theorems 4 and 1.**

Proof of Theorem 4. We use strong induction on $|S|$ (`main`). Lemma 10 gives a triangle xyz in S. If one of its edges, say uv, has $d_S(u) + d_S(v) \le |S|$, apply Lemma 11 to that edge and the third vertex. The required instance of the theorem for $S \setminus \{u, v\}$ is the induction hypothesis, since $|S \setminus \{u, v\}| < |S|$ (`card_erase_two_lt`). In Lean the three edges are handled by three calls to `case_one` with the vertices permuted. Otherwise all three degree sums exceed $|S|$, and Lemma 12 applies. □

No separate base case is needed. When $|S| \le 2$ the hypothesis fails. Inside the induction this is visible because Lemma 10 produces three distinct vertices of S.

Proof of Theorem 1. Take $V = \{0, \dots, n - 1\}$ and $S = V$. By the handshake lemma (Mathlib's `sum_degrees_eq_twice_card_edges`; Lean lemma `D_univ`), $D(V) = 2e(G) = 2(\lfloor n^2/4 \rfloor + 1)$. So Theorem 4 gives $|\mathcal{T}(V)| \ge \lfloor n/2 \rfloor$, and $\mathcal{T}(V)$ is the set of all triangles of G. □

4. Machine verification

Files and declarations. The proof is in the file

`lean/Math0/ErdosProblems/Erdos1010_erdos_1010_variants_rademacher.lean`

of the math0 repository (`https://github.com/Foundation-0/math0`), in namespace `Math0.Erdos1010`. It has 333 lines. The main theorem is stated by reference to the upstream type:

```
theorem erdos_1010_rademacher : type_of% @Erdos1010.erdos_1010.variants.rademacher := by
  intro n G h
  classical
  have key := main G _ (univ : Finset (Fin n)) rfl (by rw [D_univ, h]; simp)
  simpa [tri] using key

end Math0.Erdos1010

#math0_proves Math0.Erdos1010.erdos_1010_rademacher Erdos1010.erdos_1010.variants.rademacher
```

The auxiliary development is stated for an arbitrary `Fintype V` with decidable equality and decidable adjacency. It is instantiated at `Fin n` with the classical instances of the upstream statement. The correspondence with the text is: Definition 3: `ind`, `dg`, `D`, `tri`; Lemma 5: `dg_erase`, `D_erase`; Lemma 6: `D_erase_two`; Lemma 7: `sq_div_four`; Lemma 8: `step_bound`; Lemma 9: `exists_common` (with `dg_eq_card`); Lemma 10: `mantel` (with `exists_adj_of_D_pos`); Lemmas 11 and 12: `case_one`, `case_two` (with `mem_tri`, `tri_mono`); Theorem 4: `main`; the handshake step: `D_univ`. The proof involves no large finite computation. The only case analyses are the eight adjacency patterns in (2) and the nine pairs (i, i') in the injectivity argument. Both are discharged by `by_cases`, `fin_cases` and `simp` and are checked by the kernel like any other proof term. The file contains no `sorry`, `native_decide`, `axiom` or `set_option`.

Audit. The last command of the file, `#math0_proves`, is defined in `lean/Math0/Audit.lean`. It fails the build unless the following hold.

(1) The type of `Math0.Erdos1010.erdos_1010_rademacher` is syntactically identical to the type of the upstream declaration `Erdos1010.erdos_1010.variants.rademacher`. The comparison ignores metadata annotations only: no unfolding and no definitional equality. The universe parameters must also agree.

(2) The theorem depends on no axioms beyond `propext`, `Classical.choice` and `Quot.sound`. This is computed with Lean's `collectAxioms`. It rules out `sorryAx`, the axioms behind `native_decide`, and custom axioms.

On compilation the file reports:

```
math0 audit: 'Math0.Erdos1010.erdos_1010_rademacher' depends on axioms [Classical.choice, Quot.sound,
    propext]
math0 statement check: 'Math0.Erdos1010.erdos_1010_rademacher' proves 'Erdos1010.erdos_1010.variants.
    rademacher'
```

Toolchain and reproduction. The toolchain is Lean 4 [10] at version `v4.33.1` with Mathlib [9] at version `v4.33.1`. The upstream statements come from formal-conjectures at commit `c77fdd446d23539302aa73bcfae4a9fa962b8382` (`c77fdd4`), pinned in `lean/lakefile.toml`. To reproduce, run `scripts/verify.sh` from a checkout of the repository with `elan` installed. It fetches the Mathlib cache, builds every result with `lake build`, checks that the formal-conjectures and Mathlib package checkouts have no local modifications, and reports the number of statement checks that passed. To check this file alone, run the following in the `lean` directory. It takes well under a minute once the dependencies are built.

```
lake env lean Math0/ErdosProblems/Erdos1010_erdos_1010_variants_rademacher.lean
```

References

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[12] plby/lean-proofs, src/latest/ErdosProblems/Erdos1010.lean, theorem erdos_1010, commit 8822f7ddef30fadbd92e1c6ab4ed897af356af5e, https://github.com/plby/lean-proofs, accessed 2026-10-17.

Produced by Claude (an AI model developed by Anthropic) in the math0 project